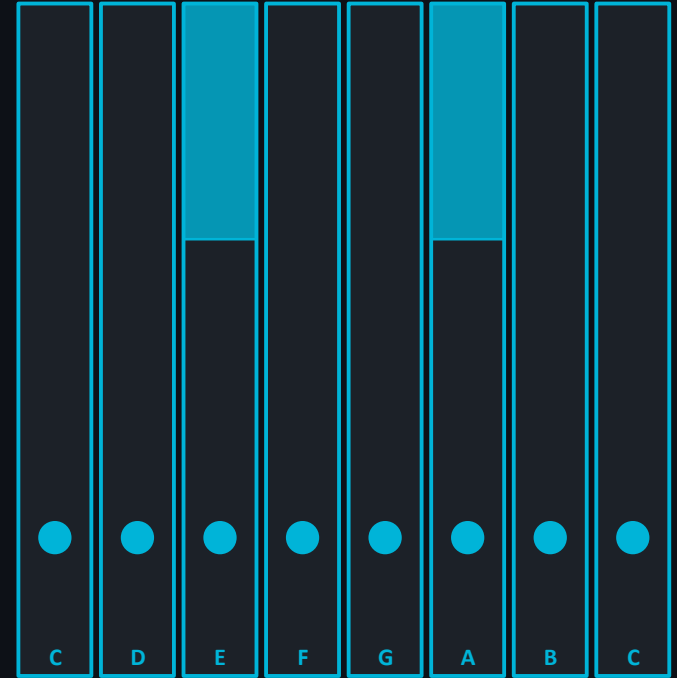




Electronic Piano Circuit

Sound generation using 555 Timer IC



Introduction



What is it?

A small circuit that produces musical notes when buttons (keys) are pressed.

Uses a 555 Timer IC to generate different sound frequencies.

Output is amplified and played through a speaker.



Why Build One?

Understand basic sound electronics.

Learn how ICs, resistors & capacitors work together.

Foundation for advanced synthesizers & keyboards.



Button Press

Switch activates



555 Timer IC

Frequency generated



Speaker

Sound produced

Objective

Core Goals

- Design and build a simple electronic piano that generates sound when keys are pressed
- Understand how ICs, resistors, and capacitors work together to produce sound
- Explore the relationship between resistor values and output frequency

Practical Applications

Music Learning

Experiment with tones and notes

Frequency Lab

Observe frequency variation visually

Synthesizer Base

Foundation for advanced keyboards

Components Required

555 Timer IC

Core tone generator

NE555 / LM555

Capacitors

Timing components

100nF + 10μF

Speaker

Audio output

8 Ohm

Power Supply

DC power input

5–9V

Resistors

Frequency control

8×1KΩ, 1×4.7KΩ

Push Buttons

Piano keys (×8)

Momentary type

Breadboard

Circuit assembly

Standard size

Jumper Wires

Connections

Various lengths

Working Principle

The 555 IC operates as an astable multivibrator, generating a square wave. Different resistor values produce different frequencies — each button connects a unique resistor, producing a unique tone.



Tone Generation

- 555 Timer in astable mode produces a continuous square wave
- Output frequency depends on resistor & capacitor values
- Formula: $f = 1.44 / ((R1 + 2 \times R2) \times C)$



Switch Activation

- Each button press connects a unique resistor to Pin 7
- This changes R2 in the timing circuit
- Result: different frequency → different musical note



Speaker Output

- Square wave from Pin 3 passes through 10μF capacitor
- Capacitor blocks DC, passes only the AC signal
- 8Ω speaker converts signal to audible sound

Assembly Steps

Part 1 of 2

01

Place IC 555

Insert 555 Timer IC on breadboard with notch facing up

02

Power Rails

Connect PIN 8 → Positive Rail, PIN 1 → Negative Rail

03

Pin Bridges

Bridge PIN 2 & 6 together; bridge PIN 4 & 8 together

04

Timing Resistor

Place 1K Ω resistor between PIN 7 and PIN 8 of the IC

05

Timing Capacitor

Place 100nF capacitor between PIN 2 and the negative rail

06

First Button

Place a momentary push button on the breadboard

07

4.7K Ω Resistor

Connect 4.7K Ω resistor between PIN 7 and one terminal of the button

Assembly Steps

Part 2 of 2

08

Connect to PIN 6

Connect other terminal of push button to PIN 6 of the IC

09

Repeat for 8 Keys

Add 7 more buttons; each with a 1K Ω resistor between its terminals (chained)

10

Output Capacitor

Place 10 μ F capacitor: (+) to PIN 3, (–) to an empty row

11

Connect Speaker

Place 8 Ω speaker between negative rail and the (–) terminal of 10 μ F capacitor

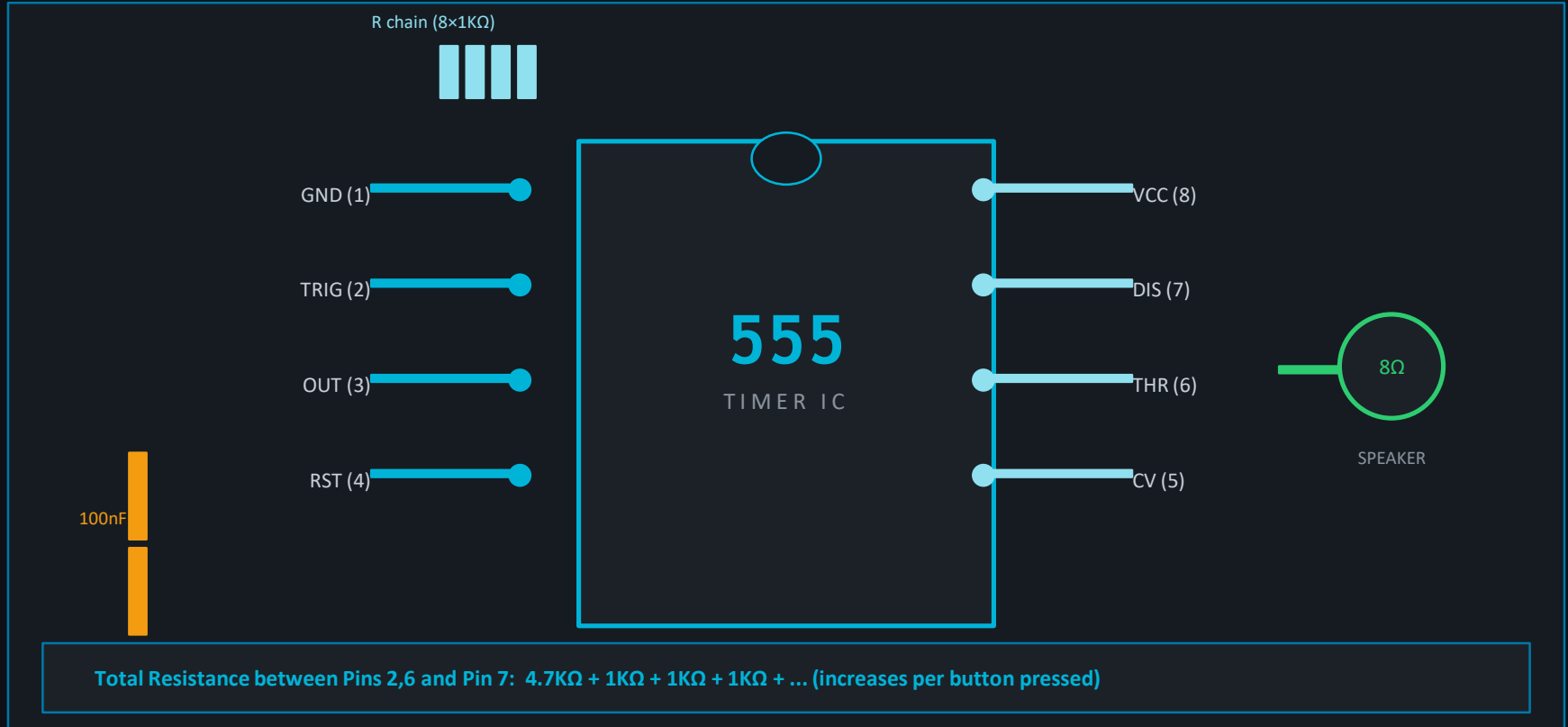
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Power Up!

Connect 5–9V DC power supply. The Electronic Piano is READY to play!

💡 Tip: Chaining buttons in series adds resistance progressively, creating 8 distinct frequencies across the octave.

Circuit Diagram



Output & Testing



Expected Results

- Each button press produces a distinct tone
- Sound is clear and continuous while key is held
- Releasing the key stops the sound immediately



Troubleshooting

No sound?

Check power supply & breadboard connections

Wrong tone?

Verify resistor values and pin connections

Weak sound?

Use a higher-power speaker or add amplifier

Note-Frequency Reference

C4

261 Hz

D4

293 Hz

E4

330 Hz

F4

349 Hz

G4

392 Hz

A4

440 Hz

B4

494 Hz

C5

523 Hz

Applications & Advantages



Toy Pianos

Musical greeting cards and toy instruments



Electronics Teaching

Hands-on sound & circuit principles



MIDI Controllers

Foundation for DIY controllers



Synthesizers

Extendable to full-scale synths

Why This Approach Shines

Low Cost

~₹100–200 in components

Simple Design

No microcontroller needed

Expandable

More keys, octaves, amplification

Educational

Real-world analog electronics

Conclusion

Summary

- The circuit uses a 555 Timer IC in astable mode to generate sound
- Resistors and buttons control the frequency of each musical note
- An 8Ω speaker converts the square wave signal into audible tones

Future Improvements

More Keys

Full musical scale (12 notes)

Arduino Integration

Digital control & MIDI output

Digital Amplifier

Better audio quality & volume



Final Thought

This project is a fun and educational gateway into analog electronics — blending music, physics, and engineering in a single breadboard circuit.



Thank You!

Electronic Piano Circuit

Questions & Discussion